PARAGON	EMITTERS (IS :13487-2024) TEST REPORT	Size :	4 - LPH
		Test Date :	02-02-2026

Date of Mfg. :	01-01-2026	**Brand Name :**	PARAGON
Batch No. :	20260101	**Type & Category :**	Unregulated, B
Mfg Qty.(No):	1957 06 29	**Nominal Test Pressure :**	1.0 kg/cm²

[1] Mechanical Tests :

Sr. No	Type of Test	Clause	Reference Value			Test Result
1.1	Construction and Workmanship	7.1	No manufacturing defects such as grooves, cracks or cavities			Satisfactory
1.2	Flow Paths	7.2	Min. Declared Flow Path=0.80			0.90
			1	2	3	
			0.85	0.89	0.95	
1.3	Resistance to Hydrostatic Pressure	7.3	No leakage shall occur through the emitter bodies or their connections to the pipe.			Satisfactory
1.4	Emitter Pull-Out	7.4	The emitter shall withstand the pulling force without pulling out the pipe wall.			Satisfactory (40 N for 1 Hour)

[2] FUNCTIONAL TESTS :

2.1	Uniformity of Emission Rate	8.1	The mean emission rate not deviate by more than 5 percent for Category A, not more than 10 percent for Category B	Mean Emission Rate : 0 LPH Deviation : 0.00%
			The coefficient of variation(Cv) of the emission rate of the test sample shall not exceed 5 percent for Category A not more than 10 percent for Category B	Cv = 0.00

Mean Emission Rate (q) = LPH

Declared Emission Rate = LPH

$$\text{Deviation from Mean Emission Rate} = \frac{\text{Declared Emission} - \text{Mean Emission}}{\text{Declared Emission}} \times 100$$

$$= \frac{\quad}{\quad} \times 100$$

VARATION =

Std. deviation of Emission Rate (Sq) =

$$\text{Co-eff. of Variation (Cv)} = \frac{Sq \times 100}{q}$$

$$CV = \frac{\quad \times 100}{\quad}$$

CV =

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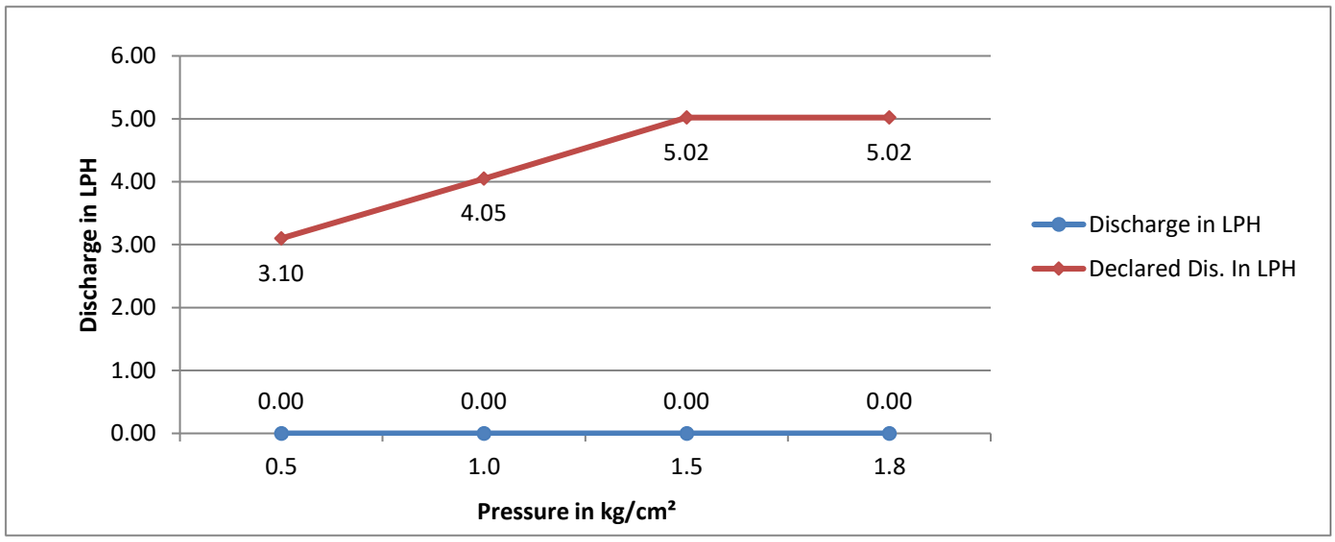
2.2	Emission Rate as Function of Inlet Pressure		8.2		The tests to be performed to determine emission rate as a function of pressure			Graph attached	
NO	**Discharge in 180 Sec. (Mililitre)**	**Discharge Ltr./ hr. (X)**	**(Mean Emission Rate Average Discharge (q) X1**	**No.**	**Ascending Order**	**(X-X1)**	**$(X-X1)^2$**	**Sq = √ $(X-X1)^2$ / 25**	**CV = (Sq /q x 100)**
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									
22									
23									
24									
25									

PARAGON	EMITTERS (IS :13487-2024) TEST REPORT	Size :	4 - LPH
		Test Date :	02-02-2026

Date of Mfg. :	01-01-2026	**Brand Name :**	PARAGON
Batch No. :	20260101	**Type & Category :**	Unregulated, B
Mfg Qty.(No):	1957 06 29	**Nominal Test Pressure :**	1.0 kg/cm²

NO	Discharge from Dripper (ml)						Discharge in LPH	DiclerDischarge in LPH	Variation %
	Pressure Kg/sq.cm	3	12	13	23	Average (ml)			
1	**0.5**					0.0	0.00	3.10	100.00
2	**1.0**					0.0	0.00	4.05	100.00
3	**1.5**					0.0	0.00	5.02	100.00
4	**1.8**					0.0	0.00	5.02	100.00



2.3	Determination of Emitter Exponent	8.3	Emitter exponent should be 0.21 to 0.5 for Unregulated, turbulent flow	m =

$$m = \frac{\sum (\log pi)(\log qi) - 1/n\,(\sum \log pi)(\sum \log qi)}{\sum (\log p_i)^2 - 1/n\,(\sum \log p_i)^2}$$

q = emission rate, in l/h;
k = constant;
p = inlet pressure, in Kpa;

m= emitting unit exponent;
I = 1,2,3,4,…….,n;
n = number of pressure value used in 8.2

No	p Kg/cm²	pi Kpa	qi	pi x qi	(pi)² Kg/cm²	(pi)² Kpa
1						
2						
3						
Total						
Log =						
(log pi)² =						

$$m = \frac{\sum (\log pi)(\log qi) - 1/n\,(\sum \log pi)(\sum \log qi)}{\sum (\log p_i)^2 - 1/n\,(\sum \log p_i)^2}$$

m = _____

m= _____

m= _____

m=